

SEARLE FIELD, NE, USA

WMO: 725621

Lat: 41.119N

Lon: 101.768W

Elev: 991

StdP: 89.98

Time Zone: -7.00 (NAM)

Period: 97-19

WBAN: 94063

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-18.6	-15.9	-22.3	0.6	-17.8	-19.1	0.8	-14.3	15.2	0.6	13.6	1.0	2.8	260	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	36.2	19.5	34.1	19.4	32.6	19.1	22.2	30.1	21.5	29.5	20.7	28.7	5.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	16.7	25.1	19.1	15.7	24.6	18.1	14.7	23.8	70.5	30.2	67.3	29.3	64.6	28.8	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.4	9.7	-25.3	38.8	2.2	1.7	-26.9	40.0	-28.1	41.0	-29.4	42.0	-30.9	43.3		
(5)			-25.6	23.9	2.1	1.1	-27.1	24.7	-28.3	25.4	-29.5	26.0	-31.1	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-2.1	-1.1	4.4	8.9	14.4	20.7	24.2	22.5	17.9	9.9	3.3	-2.2		
	DBStd	10.55	5.72	6.01	5.59	5.01	4.77	3.95	2.88	3.03	4.54	5.06	5.27	5.68		
	HDD10.0	1608	374	310	184	78	15	1	0	0	2	62	204	378		
	HDD18.3	3468	632	543	432	284	139	21	2	6	62	262	449	636		
	CDD10.0	1663	0	1	11	44	151	321	442	388	241	60	4	0		
	CDD18.3	482	0	0	0	1	17	92	185	136	50	2	0	0		
	CDH23.3	6271	0	1	25	75	338	1193	2209	1521	779	124	5	0		
(13)	CDH26.7	2908	0	0	3	14	117	533	1152	713	343	33	0	0		
(14)	Wind	WSAvg	3.9	3.8	4.1	4.4	5.0	4.4	4.0	3.3	3.1	3.4	3.7	3.8		
(15) Precipitation	PrecAvg	495	9	13	25	56	76	83	76	56	42	36	15	8		
	PrecMax	670	27	46	64	114	174	198	143	128	121	81	55	41		
	PrecMin	176	0	1	2	0	19	0	24	15	8	3	0	0		
	PrecStd	102	7	10	19	33	40	45	29	30	24	23	14	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	19.9	27.0	28.6	32.8	37.4	38.6	37.2	36.0	30.2	23.7	17.0		
		MCWB	6.8	8.5	11.5	13.3	16.4	18.5	19.2	19.2	17.8	14.5	10.8	7.5		
	2%	DB	12.9	15.4	22.5	25.9	30.0	34.5	36.8	34.9	33.0	27.1	20.0	12.6		
		MCWB	5.1	6.1	9.8	12.2	16.0	19.0	20.1	19.5	17.6	13.8	9.4	5.2		
	5%	DB	9.9	12.1	18.7	22.5	27.3	32.3	34.9	32.8	31.0	23.6	17.0	9.3		
		MCWB	3.9	5.0	8.8	11.4	15.3	18.5	20.0	19.4	17.3	12.6	8.1	3.5		
	10%	DB	7.0	8.5	15.2	19.1	24.4	29.8	32.7	31.0	27.9	20.2	12.8	6.5		
		MCWB	2.2	3.2	7.1	10.2	14.4	18.3	20.0	19.4	16.6	11.2	6.2	2.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	8.8	13.1	14.9	19.2	22.4	23.6	23.3	21.1	16.7	11.9	7.9		
		MCDB	15.0	17.6	21.6	23.4	27.4	29.7	31.1	30.5	28.8	26.0	22.2	15.7		
	2%	WB	5.5	6.8	11.0	13.2	17.7	21.0	22.4	21.9	19.8	14.8	9.9	5.5		
		MCDB	12.7	14.3	20.3	21.9	25.2	28.9	30.7	29.4	27.7	22.9	19.3	11.9		
	5%	WB	4.0	5.1	9.4	12.0	16.6	20.2	21.7	21.2	18.8	13.4	8.3	3.8		
		MCDB	9.1	11.6	17.6	20.9	24.2	28.3	30.2	28.6	27.0	21.7	16.0	9.1		
	10%	WB	2.5	3.4	7.8	10.8	15.4	19.2	21.0	20.4	17.8	12.1	6.6	2.1		
		MCDB	6.6	8.3	15.0	18.4	22.8	27.4	29.4	27.6	25.5	19.1	12.8	6.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	14.0	15.7	14.7	14.5	15.0	15.3	15.0	16.5	16.2	15.8	13.9		
		MCDBR	19.4	20.3	23.2	21.4	19.3	18.9	19.0	18.6	20.8	22.8	22.1	18.7		
		MCWBR	12.3	12.1	12.0	10.0	8.2	6.4	6.0	6.5	7.7	10.6	12.3	12.0		
	5% WB	MCDBR	18.0	19.4	20.8	18.8	15.8	15.7	15.6	15.2	16.6	19.5	21.3	17.8		
		MCWBR	11.8	11.9	11.3	9.4	7.4	6.6	5.9	6.3	7.1	9.8	12.1	11.7		
(40) Clear-Sky Solar Irradiance	taub		0.257	0.265	0.290	0.315	0.341	0.356	0.381	0.372	0.335	0.295	0.266	0.257		
	taud		2.554	2.531	2.486	2.432	2.432	2.419	2.393	2.404	2.472	2.554	2.557	2.535		
	Ebn at Noon		925	965	968	957	933	916	889	889	904	914	904	894		
	Edh at Noon		69	82	97	110	113	115	117	112	98	80	68	65		
(44) All-Sky Solar Radiation	RadAvg		2.35	3.20	4.41	5.43	6.30	7.26	7.16	6.23	5.07	3.57	2.55	2.01		
	RadStd		0.16	0.20	0.32	0.30	0.45	0.41	0.26	0.27	0.22	0.32	0.15	0.12		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page